ACADEMIC PROJECT PRESENTATION

AI-Powered Credit Risk Intelligence Platform

Explainable Credit Risk Decision Support with ML, EDA & Natural-Language Analytics

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Class Imbalance | TreeSHAP Explainability | Conversational Analytics

Use Case

PLATFORM OBJECTIVE

“Design and build a lightweight AI-powered credit risk platform that combines data analysis, machine learning, explainability and conversational analytics.”

USE CASE SCOPE

1

Credit Risk Assessment:

Using the Home Credit Default Risk dataset (307k+ loan applicants).

2

Pattern & Insight Discovery:

Identify relationships from applicant, credit, and repayment history.

3

Predictive Default Modeling:

Train and evaluate an ML model to predict credit default risk.

4

Explainable Predictions:

Provide transparent, applicant-level feature attributions.

5

Natural-Language Analytics:

Enable conversational portfolio exploration via Talk-to-Data.

AI ENGINEERING MODULES

1

Data Understanding & EDA

Data quality, distributions & 5+ key business insights

2

Machine Learning Layer

Class imbalance handling, risk score & tier classification

3

Explainable AI (XAI)

TreeSHAP local feature attribution for decision transparency

4

Talk-to-Data System

NL-to-SQL query engine with safety validation & plain-English answers

5

User Interface & Docker

Lightweight multi-section web UI & one-command containerization

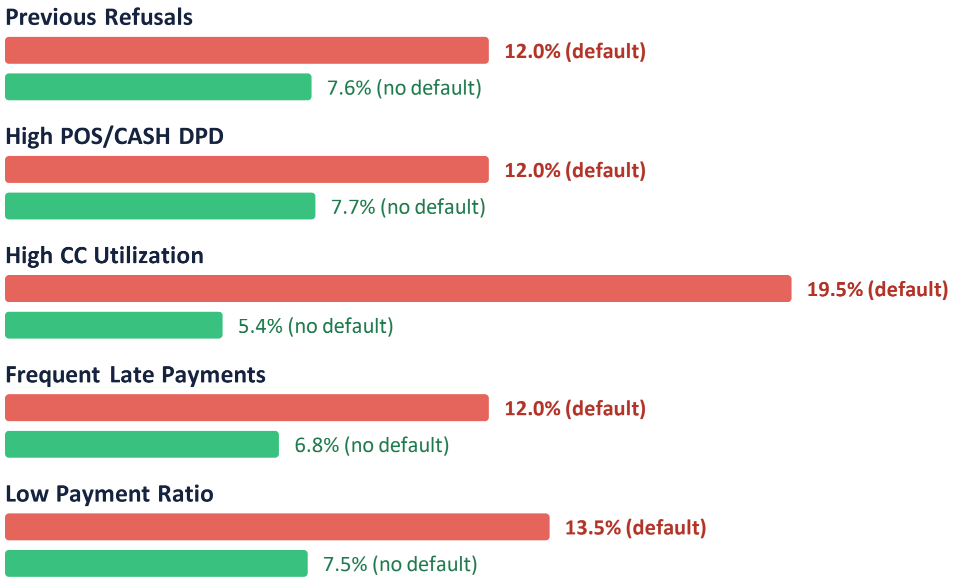
Dataset & Key EDA Findings

307,511
TOTAL APPLICANTS

24,825
DEFAULTS (TARGET = 1)

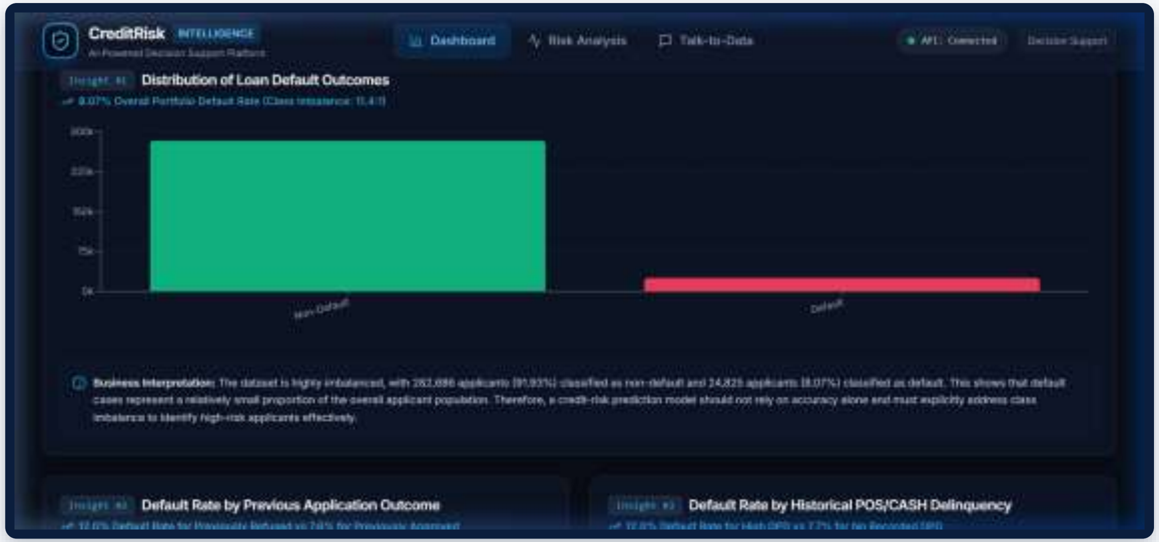
8.07%
BASELINE DEFAULT RATE

Key EDA Insights — Observed Default Rates



**Observed historical associations — not causal.*

Portfolio EDA Dashboard



Live application screenshot — default outcome distribution & business interpretation

Machine Learning Approach & Model Selection

Raw Data › Feature Engineering › 231 Features › Stratified Split › Class Weighting › Model Benchmarking › CatBoost Threshold 0.66

Model Comparison (Validation Set N = 61,503)

Model	ROC-AUC	PR-AUC
Logistic Regression	0.7735	0.2608
XGBoost	0.7874	0.2848
LightGBM	0.7870	0.2844
CatBoost ✓ (Selected)	**0.7878**	**0.2862**

PR-AUC is the primary metric — chosen for severe class imbalance (8.07% defaults).

Why CatBoost?

- **Highest PR-AUC:**
  0.2862 — best at identifying minority defaults.
- **Highest ROC-AUC:**
  0.7878 — best overall ranking discrimination.
- **Native Categorical:**
  Handles high-cardinality variables without one-hot explosion.
- **Operating Threshold:**
  Tuned to 0.66 on validation set to maximise F1 score.

Class Distribution (Validation Set)

Non-default: 91.93% | Default: 8.07%

Final Model Results & Risk Scoring

0.7878	0.2862	0.2824	0.4354	0.3426	0.8651
ROC-AUC	PR-AUC	PRECISION	RECALL	F1 SCORE	ACCURACY

Confusion Matrix (N = 61,503)

Threshold = 0.66

	Pred 0 (Non-default)	Pred 1 (Default)
Actual 0	51,044 TN	5,494 FP
Actual 1	2,803 FN	2,162 TP ✓

Tuned on validation data to maximise F1.

43.5%	9.7%	28.2%	0.343
True Positive Rate	**False Positive Rate**	**Precision**	**F1 Score**
Of actual defaults correctly flagged	Non-defaults incorrectly flagged	Flagged applicants that truly default	Harmonic mean of precision & recall

Risk Score = Probability × 100

LOW	< 30	Approve / Low Risk
MEDIUM	30 – <60	Manual Review Required
HIGH	≥ 60	High Risk / Reject Recommendation

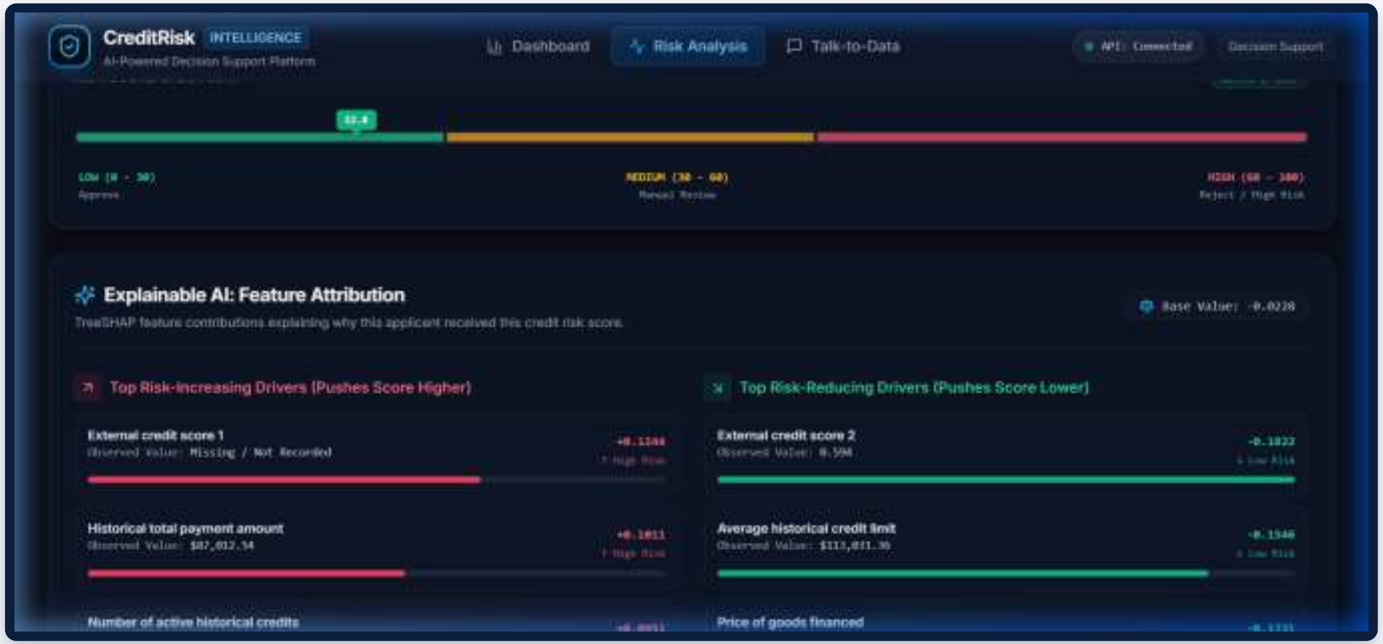
Verified Benchmark Applicants

ID 396899	0.228 → 22.83	LOW	Approve
ID 100002	0.828 → 82.76	HIGH	Reject Recommendation

**Decision-support — not autonomous lending.*

Explainable AI with SHAP

SHAP Waterfall — Applicant 396899 (Live Application Output)



SHAP EXPLANATION

Base Value (Expected log-odds):
-0.0228 (corresponds to ~49.4% base rate logit).

Model Output (Applicant 396899):
Default probability 0.2282 (Risk Score: 22.83 / Low Risk).

Top Positive Driver (+Risk):
TOTAL_PAYMENT_AMOUNT (+0.1244) & AMT_ANNUITY (+0.1011).

Top Negative Driver (-Risk):
EXT_SOURCE_2 (-0.1822) & BUREAU_AVG_LIMIT (-0.1546).

TreeSHAP Explainability

- **Local Attribution:**
  Exact applicant-level contributions computed per prediction.
- **Positive value:**
  Feature increases predicted default risk.
- **Negative value:**
  Feature reduces predicted default risk.
- **Fair-Lending Guardrail:**
  Protected demographic attributes are excluded from user-facing individual explanations.

Applicant 396899 — Key Drivers

↑ Risk-Increasing

- TOTAL_PAYMENT_AMOUNT
- BUREAU_ACTIVE_COUNT
- AMT_ANNUITY

↓ Risk-Reducing

- EXT_SOURCE_2
- BUREAU_AVG_LIMIT
- AMT_GOODS_PRICE

**SHAP = model contribution, not real-world causation.*

Talk-to-Data: Natural-Language Analytics

NL → SQL Flow

User Question
▼
Safety Check
▼
Deterministic / LLM
▼
AST SQL Validator
▼
Read-Only SQLite
▼
Grounded Answer

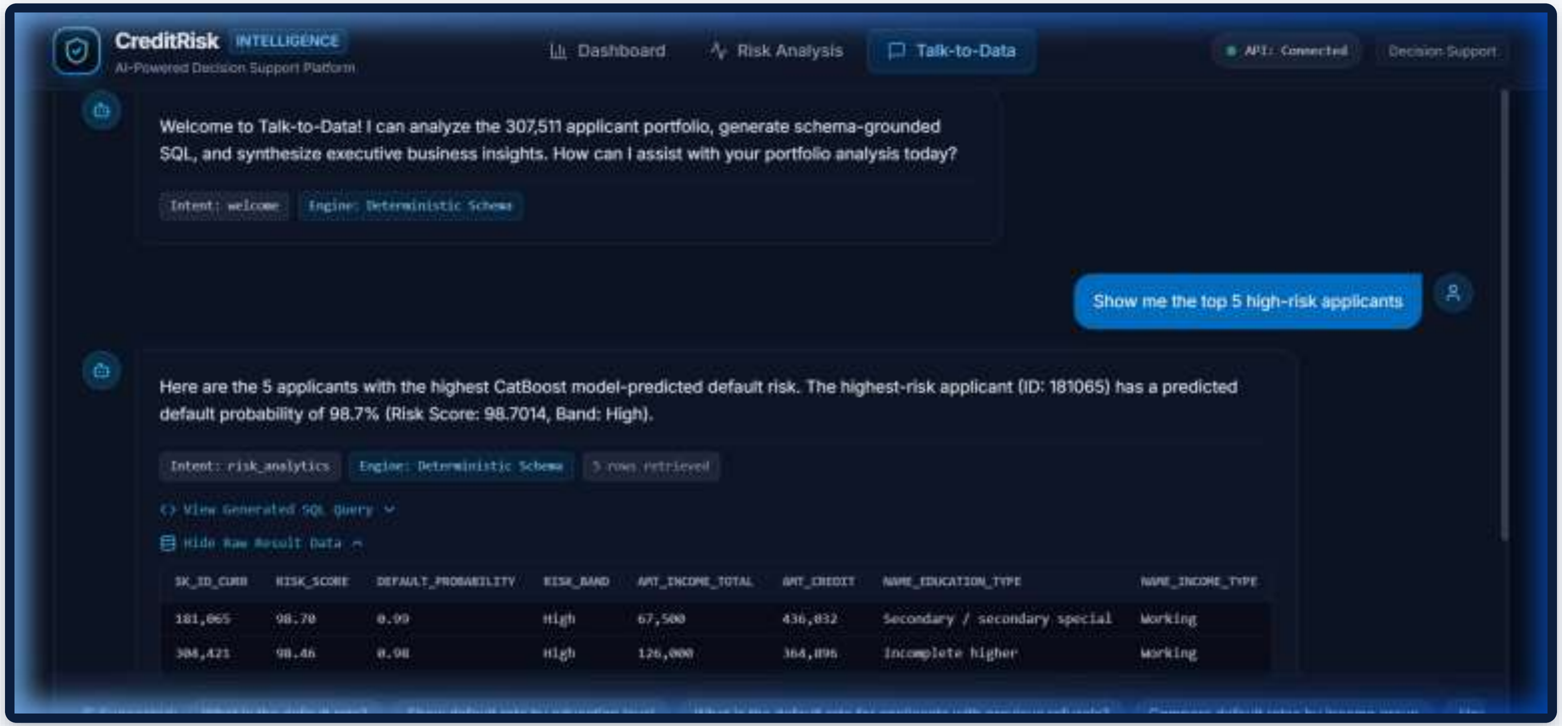
Example Questions

"What is the default rate?"

"How many applicants have income > 300,000?"

"Show top 5 high-risk applicants."

Live Talk-to-Data Query — Top 5 High-Risk Applicants



Deterministic fast-path:

Intent matching without LLM for speed & reliability.

Schema-aware LLM path:

LLM generates SQL aware of actual table schema.

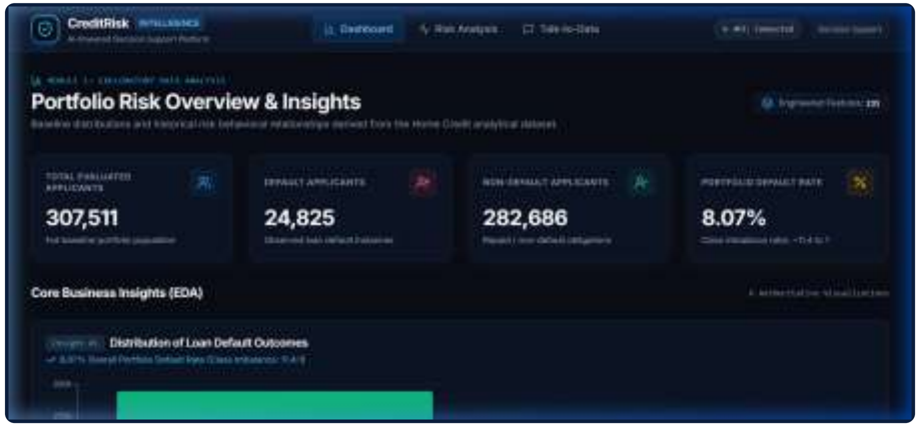
AST safety check:

All SQL validated via abstract syntax tree before execution.

Read-only SQLite:

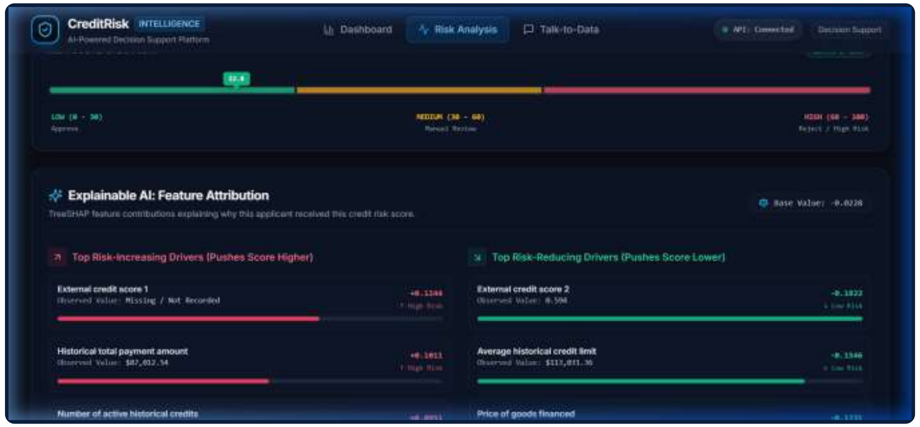
No write operations possible — analytics DB is protected.

Application Outputs

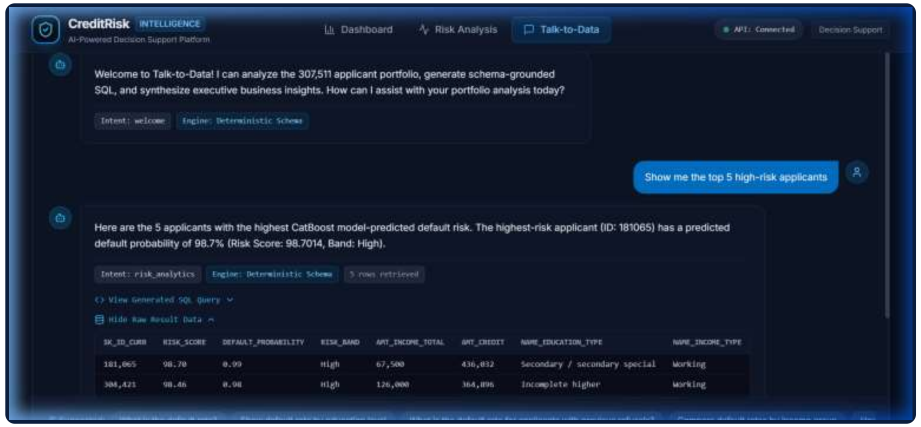


Portfolio EDA

CatBoost Risk Analysis



SHAP Explanation



Natural-Language Analytics

Design Decisions, Limitations & Conclusion

Major Design Decisions

- **CatBoost + PR-AUC:**
  Highest PR-AUC (0.2862) on 8.07% imbalanced class.
- **Threshold 0.66:**
  Tuned on validation set to maximise F1 score.
- **Dual SQLite Stores:**
  Analytical + inference DBs — eliminates 2.64 GB raw CSV at runtime.
- **Pre-scored Risk Table:**
  All 307,511 applicants scored for instant portfolio queries.
- **Hybrid NL-to-SQL:**
  Deterministic fast path + schema-aware LLM for safe analytics.
- **TreeSHAP:**
  Local attribution with demographic masking as a fair-lending guardrail.

Limitations

- **Historical Dataset:**
  Static snapshot; model behaviour on new distributions not tested.
- **Single Split:**
  One stratified split used; k-fold cross-validation not performed.
- **Moderate PR-AUC:**
  0.2862 reflects inherent difficulty of rare-event default prediction.
- **No Calibration:**
  Probability calibration (Platt / Isotonic) not yet applied.
- **Fairness Audit:**
  Formal statistical fairness tests not yet implemented.
- **LLM Dependency:**
  Conversational queries need LLM uptime; deterministic fallback available.

Future Work & Ethics

- **Prob. Calibration:**
  Platt scaling or isotonic regression for well-calibrated scores.
- **Fairness Testing:**
  Statistical parity, equalised odds across demographic groups.
- **Cross-Validation:**
  k-fold evaluation for more robust generalisation estimates.
- **Drift Detection:**
  Flag feature distribution shift on incoming applicant batches.
- **Prototype Status:**
  Academic decision-support tool — not a production lending system.

"EDA + Explainable ML + Natural-Language Analytics — one credit-risk decision-support platform."

AI-assisted decision support — not autonomous lending.